

PAPER_1498 — U-UA Coupling Constant = $1/SO_5^4 = 10^{-4}$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Millennium) / Cross-domain identity **Date:** June 17, 2026 **Status:** CLOSED — Millennium Prize UQFF coordinator constant matches Sun quiet B-field identity

Observation

PAPER_563 (Millennium Prize UQFF Coordinator — All Six Problems) lists the canonical UQFF calibrated constants: - $\kappa = 5.0e-4 \text{ day}^{-1}$ - $[SSq] = 5.7e-1$ - $H_{SCm} \approx 9.9e-1$ - **U-UA $\approx 1.0e-4$** ← target - $k_\eta = 1.0e-113$ - $\beta_i \approx 6.0e-1$

UQFF Closed Identity

$$U_{UA} = 1 / SO_5^4 = 1 / 10^4 = 1.0 \times 10^{-4} \quad \text{EXACT}$$

Cross-Domain Pairing

This identity is structurally identical (in dimensionless form) to: - **Sun quiet B-field** = $1/SO_5^4 = 10^{-4} \text{ T}$ (PAPER_1486 / F:/Aetheric Propulsion)

Same integer-primitive form, different physical domains (coupling vs. magnetic field), different units. The shared identity reveals SO_5^4 as a recurring suppression factor across UQFF coupling and field hierarchies.

NOT REPLACEMENT

SM has no analogue — U-UA is the UA-channel coupling specific to UQFF. The value 10^{-4} in SM appears only as fitted dimensionless ratios (e.g., gauge-coupling running) with no structural origin.

Reference

- Source: PAPER_563 Millennium Prize UQFF Coordinator
- Related: PAPER_1486 (Sun quiet B-field), all 6 Millennium derivations (Bucket A)
- Calculator dispatch: `calculate_paradox({"paradox": "u_ua_coupling_constant"})`

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